
Algorithm : Sequential Checksum & Early-Exit Truncation

Input : model M , internal classifiers $IC_1 \dots IC_N$, sample x

Output : Final prediction y

for $k = 1$ **to** N **do**

Step 1 : Compute Runtime Checksum

$H_k \leftarrow \text{compute_sha256}(M.\text{block}_k.\text{parameters})$

Step 2 : Validate Integrity

if $H_k \neq \text{Original_}H_k$ **then**

$\text{SafePool} \leftarrow \{IC_i \mid i < \text{first_IC_in}(\text{block}_k)\}$

break

Step 3 : Stochastic Branch Selection

$\text{Selected_ICs} \leftarrow \text{random_sample}(\text{SafePool}, \text{size} = q)$

Step 4 : Multi-Branch Logit Generation

$\text{Logits} \leftarrow []$

for IC_j **in** Selected_ICs **do**

$\text{Logits.append}(\text{forward_pass}(x, IC_j))$

Step 5 : Confidence-Weighted Consensus

$\text{Probs} \leftarrow []$

for l **in** Logits **do**

$\text{Probs.append}(\text{softmax}(l))$

$y \leftarrow \arg \max \left(\frac{1}{q} \sum_{p \in \text{Probs}} p \right)$
